

Unifying Multi-Warehouse Pharma Sales Data

Entity Resolution, Impact Analysis, ML & Operational HITL — Final Report

Dataset: 10,000 sales transactions from 10 independent pharmaceutical warehouses (Egypt) · **Pipeline:** 4 notebooks + Streamlit dashboard + live n8n HITL system · **Date:** June 2026

1. Executive summary

Ten warehouses recorded the same medicines, pharmacies, and sales reps under different names, with no shared keys. This report describes a fully local pipeline that resolves those entities safely, models the result as a star schema, and quantifies the payoff: **28.9% of total revenue (EGP 747,907) was fragmented across name variants** and invisible to any per-name analysis. After resolution, 5,045 raw product names collapse to 4,370 master products, the true top customers and rep leaderboard emerge, and the remaining genuinely ambiguous cases — 735 pairs, revenue-impact sorted — are routed to a human-in-the-loop review queue served live through Telegram via two validated n8n workflows.

The architectural thesis throughout: **the model decides, rules guard, and humans resolve what neither can**. In pharmaceutical data a false merge (Brufen 200mg absorbed into 400mg) corrupts medical reality, so every layer is built to make false merges structurally impossible rather than merely improbable, and every accuracy claim in this report is measured, not asserted.

2. Data reality and cleaning (Notebook 1)

Profiling established constraints that shaped everything downstream. The four geography columns are 100% null and were dropped (stakeholder-confirmed). Identifiers are supplier-scoped: the same `prod_id` or `account_id` means different things at different warehouses, which is the core problem motivating entity resolution. Export windows are wildly disjoint — supplier 60's entire extract covers a single day, supplier 69 two days, supplier 76 is 2022–2024 history — so any cross-supplier time-series claim is only valid inside overlapping windows, and aggregate monthly “trends” are export artifacts.

Cleaning was deliberately conservative: 10 rows with corrupted dates (0000-00-00) quarantined rather than deleted, one zero-quantity/zero-amount non-transaction removed, and 33 statistical outliers **flagged, not deleted** — large orders are plausible in B2B distribution, and the two largest “duplicate-looking” transactions turned out to be legitimate repeat orders on different invoices. Final working set: 9,989 transactions with a full cleaning ledger in the QA report.

3. Entity resolution (Notebook 2) — the core

3.1 Why plain fuzzy matching is unsafe here

Arabic orthographic variance (ا/إ, ي/ي, ه/ه, spacing, glued digits like 14مستات40) breaks edit-distance similarity in one direction, while the opposite failure is worse: **145 candidate pairs in this dataset have string similarity above 0.85 but conflicting strengths** — بروفين 200/400/600 score as near-duplicates. A matcher driven by string similarity alone would commit those merges. This single measured number justifies the layered design.

3.2 The pipeline

Each entity type passes through the same architecture, tuned per entity:

Normalization unifies Arabic orthography, strips noise tokens (the س ج / س-ج / س ق price markers), splits glued letter-digit boundaries, and converts Eastern Arabic numerals. **Attribute extraction** then parses each product name into structured fields — base name, strength, form, pack size, size marker (كبير/صغير), manufacturer — converting a string-matching problem into a structured one. Strength and form become *hard constraints* the model can never override. A fallback captures bare numbers with no unit (180) as strengths with unspecified units, compared through a numeric-aware comparator that treats 180 = 180mg as agreement, 180 vs 120 as conflict, and a bare stage number vs a pack weight (1 جم/400) as incomparable rather than conflicting.

Candidate generation uses character 2–4-gram TF-IDF with cosine nearest neighbours — language-agnostic, robust to Arabic spelling variance, and fully local. It reduced 12.7 million potential comparisons to 10,685 candidate pairs (a ~1,190× reduction) with no semantic model downloads, no external APIs, and no data leaving the environment.

The pairwise classifier is a Random Forest over string-similarity and attribute-agreement features. Because no labels exist on day one, it is bootstrapped by **weak supervision**: unambiguous positives (near-identical or word-reordered names with compatible attributes; 324 pairs) and unambiguous negatives (hard conflicts and low-similarity pairs; 5,949) are generated by rules, the model interpolates the gray zone, and HITL decisions progressively replace the weak labels — *weak supervision, human-refined*. Cross-validated precision and recall on the weak labels are ≈0.99, and tier thresholds were chosen from a threshold sweep, not assumed.

Tiering grants auto-merge bars graduated by strength evidence, always requiring zero soft conflicts (form, size marker, manufacturer, pack): explicit strength agreement earns the lowest bar; pairs with no strength on either side need either a high model score or exact word-reorder equivalence; and pairs with strength missing on one side can auto-merge only through the **uniqueness bypass** — when the base name has exactly one strength anywhere in the catalog, ambiguity is provably impossible (يوريبيان exists in one strength; plain “يوريبيان شراب” can be nothing exists in three strengths, no similarity score justifies guessing, and the pair goes to humans. A rule-based safety net additionally routes high-lexical-overlap pairs that the model under-scores into the review queue: rules catch what the model is blind to.

Clustering runs connected components over auto-merge edges only, then applies two guards: no cluster may contain two different strengths (this kills chain-merging across drug families by construction), and oversized clusters are dissolved into review rather than trusted. In the final run the strength guard fired 32 times — the second line of defense doing real work after the lexical gates widened.

3.3 What adversarial self-audit caught

Two real bugs were found by auditing the pipeline’s own output and are documented as design rules. First, the safety net’s compatibility condition silently excluded missing-strength pairs — the exact class it existed to protect. Second, in the lighter rep configuration, a bare first name (“MOSTAFA”) subset-matched every “مصطفى X” and connected components chained four different reps into one person; the fix requires *sort* (not token-set) similarity for auto-merge and forbids single-token names from ever auto-merging. The same subset trap was closed for pharmacies (ص حبيب فيصل vs ص ريمون حبيب فيصل). The matcher is trusted because it was attacked, not because it was assumed correct. A third € followed the same discipline: supplier shorthand for pack units («ش4» for 4 «ك14» كيسول شوكت), was invisible to the original pack regex, leaving pack sizes unextracted and comparison strings inconsistent. Expanding the pattern with domain abbreviations *and* canonicalizing the shorthand inside the normalized string (ش4ش14) — plus stripping the extracted manufacturer suffix from lexical comparison — raised auto-merges from 477 to 675 and lifted the measured fragmented-revenue figure from 22.3% to 28.9%: fixing extraction upstream proved safer and more effective than loosening any confidence threshold downstream.

3.4 Results and the review queue

Entity	Raw entities	Masters	Variants merged	HITL pairs
Products	5,045 names	4,370	675	521
Pharmacies	1,103 accounts	1,100	3 cross-supplier	89
Sales reps	136 entries	108 + UNKNOWN	28	81

The product queue was reduced from an initial 885 pairs to 565 (–36%), while auto-merges rose 477 → 675 by the uniqueness bypass (184 pairs), lexical gating for strength-matching pairs, and word-reorder clearance — **without relaxing any medical constraint**. The remaining queue is dominated by subset pairs with no strength on either side (بانادول كولد اند فلو vs بانادول كولد), which are undecidable by any rule and below design. The queue is sorted by revenue at stake, so the first ~50 reviews resolve the bulk of the financial uncertainty; the top pair alone covers EGP 80K. Names such as اي سيلاي and كلاستر were identified as channel/system accounts rather than people and tagged accordingly.

3.5 Price-based disambiguation — a hypothesis, measured

A stakeholder-proposed enhancement — resolving missing-strength items by matching unit price against known variants in the same time window — was tested rather than adopted on faith. Three findings: gross unit price (discount backed out) is effectively frozen within product, supplier, and month (median CV ≈ 0.000), so the signal is real; strengths are separable (median 37% price gap, 77% of groups above 15%); but a holdout experiment that hid known strengths and tried to recover them from *other suppliers’* prices achieved only **64% precision**, partly because box price conflates strength with pack size. The design conclusion grants price three roles by evidence strength: same-supplier match auto-resolves (deterministic but ~1–2% coverage on this snapshot, growing with live history); cross-supplier match becomes a **pre-selected suggestion in the HITL review, never a decision**; and price far outside a variant’s band serves as reliable negative evidence for pruning. Sixty-five queue pairs now carry price suggestions.

4. Modeling and measured impact (Notebook 3)

The warehouse is a standard star schema — FactSales at invoice-line grain against Product, Pharmacy, Rep, Supplier, and Date dimensions — with original supplier-scoped keys retained on every fact row, so any unified number traces back to warehouse-level lines.

The before/after comparison quantifies the ROI. **474 master products existed under 1,149 different names, hiding EGP 747,907 — 28.9% of all revenue — from any per-name analysis.** The most fragmented product, 20 كيسول, split across four names whose best individual rank was #2,747; unified, it ranks #995 — a climb of 1,752 positions. The unified customer view exposes pharmacies buying from multiple suppliers that every per-warehouse report undercounted. On the rep side, **EGP 183,660 of “rep” revenue belongs to channel/system accounts** and is excluded from the people leaderboard, while EGP 82,212 is honestly reported as unattributable rather than guessed.

5. Machine learning (Notebook 4)

ML deliberately comes last: forecasting fragmented entities means forecasting noise.

Forecasting is scoped to the seven suppliers with at least 120 days of continuous coverage and to weekly grain, where 27 supplier×product series have ten or more active weeks. On a four-week holdout the benchmark reads naive 1.12, four-week moving average 1.05, damped ETS 0.90 (mean WMAPE) — ETS wins on average but not per series, so the deliverable is a **champion/challenger harness** that ships whichever model currently beats the baselines per series, re-evaluated weekly. With 25–70 points per series, classical methods are the honest ceiling; anything deeper would be theater on this snapshot.

Anomaly detection runs three lenses: an IsolationForest over amount, quantity, discount, and *deviation from the master product’s own median price* — a feature that only exists because of entity resolution — flagging the top 1% of transactions (100); robust z-scores on weekly revenue per supplier producing 33 drop/spike alerts (the future Telegram alert feed); and rep-behaviour screens that surfaced three reps whose discount policy deviates more than two robust-z from peers plus one revenue-collapse week.

6. Dashboard

An eight-page Streamlit application reads the warehouse directly: Overview (with supplier coverage Gantt making the disjoint export windows impossible to miss), Unification Impact (the before/after ROI page), Products, Pharmacies, Sales Reps (person/channel/unknown split), Anomalies & Alerts, Forecast Bench, and the HITL Queue with revenue at stake and price suggestions. Trend charts are drawn per supplier with an explanatory caption rather than as a misleading unified line. The app ships as a self-contained bundle with a Dockerfile for deployment.

7. Operational implementation: n8n + Telegram HITL (built and deployed)

The HITL design was implemented on a live n8n instance, not left as a diagram. Three **n8n Data Tables** act as the operational store: `hitl_review_queue` (uncertain pairs with model score, revenue at stake, price suggestion, decision, and status — pre-sorted by financial impact), `supplier_week_alerts` (the 33 anomaly alerts with robust z-scores and a sent flag), and `kpi_summary` (digest metrics). Two workflows were built and validated with zero errors:

“Pharma HITL Review — Telegram” turns review into a tap-tap-tap flow. A Telegram trigger receives both chat commands and button taps; a normalization step distinguishes them. On a tap, the workflow acknowledges the button, writes the decision (MERGE/SEPARATE/SKIP) back to the queue table, marks the pair done, fetches the next pending pair, and sends a new review card — both raw names, the model’s confidence, the revenue at stake, and, where the price layer produced one, a pre-computed suggestion — with inline Merge / Separate / Skip buttons. Each decision automatically serves the next pair; an empty queue is announced explicitly.

“Pharma Daily Digest & Alerts — Telegram” runs every morning at 08:00: it reads the KPI table, counts pending reviews, pulls unsent anomaly alerts, composes a single digest (totals, top product/rep/pharmacy/supplier, queue depth, and the top-5 alerts by |z| marked ▼DROP/▲SPIKE), sends it, and marks the alerts sent so none is re-broadcast.

A `sync_to_n8n.py` script closes the production loop: after each pipeline re-run it refreshes all three tables, replacing pending pairs while **preserving every human decision** — and those accumulated decisions are precisely the labeled data that retrains the pairwise classifier, completing the weak-supervision → human-refined cycle. The one-time historical backlog is ~150 high-impact pairs; daily delta runs are expected to add a handful.

8. Limitations and future work

The snapshot’s disjoint export windows limit trend and forecast claims to within-supplier windows; live continuous data removes this constraint and simultaneously grows the coverage of deterministic same-supplier price resolution. Auto-merge precision is calibrated on weak labels and audited samples; the exported 80-pair audit file and accumulating HITL decisions provide the human-labeled ground truth for formal precision reporting and retraining. Pack-size extraction (32% coverage) can be deepened to enable per-tablet price normalization, which would directly improve cross-supplier price transferability. Pharmacy resolution is intentionally conservative (three auto-merges) given name-collision risk; address enrichment would unlock more. Finally, multilingual embeddings remain a candidate additional feature for the pairwise classifier — evaluated against, not assumed better than, the char-n-gram baseline that proved sufficient here.

Deliverables: 01_profiling_cleaning.ipynb · 02_entity_resolution.ipynb · 03_modeling_impact.ipynb · 04_ml.ipynb · normalize.py · Streamlit dashboard bundle · warehouse/ star schema · hitl_queue.csv · audit_sample.csv · this report.